

Year 2 Assessed Problems

Semester 2

Assessed Problems Sheets

SOLUTIONS TO BE SUBMITTED

ON CANVAS BY

Wednesday 18 February 2026 at 17:00

Assessed Problems 2

Richard Mason

Problem 1. A set of vectors $\{\hat{e}_1, \hat{e}_2, \hat{e}_3\}$ form an orthonormal basis with respect to the standard complex inner product

$$(\mathbf{x}, \mathbf{y}) = \sum_{n=1}^3 x_n^* y_n$$

If

$$\begin{aligned} \mathbf{f}_1 &= \hat{e}_1 - \hat{e}_2 \\ \mathbf{f}_2 &= \hat{e}_1 + \hat{e}_2 \\ \mathbf{f}_3 &= \hat{e}_3 \end{aligned}$$

1. Show that these vectors form an orthogonal system. [3]
2. Normalise the vectors. [2]
3. If $\mathbf{s} = \hat{e}_1 + i\hat{e}_2 + \hat{e}_3$, and $\mathbf{t} = \hat{e}_1 + 2\hat{e}_2$ express these in terms of your orthonormal basis $\{\hat{\mathbf{f}}_1, \hat{\mathbf{f}}_2, \hat{\mathbf{f}}_3\}$. [4]
4. Is $(\mathbf{s}, \mathbf{t})$ is the same in both bases. [1]

Structure in the Universe (03 00554)

Problem Sheet 3

A main-sequence star of mass $10.0 M_{\odot}$ (i.e. 10 solar masses), radius $4.0 R_{\odot}$ (i.e., four solar radii), and rotational period 10 days collapses to form a pulsar of radius 11 km. Estimate, stating any assumptions made:

- a) The rotational period of the pulsar **[4]**
- b) The rotational kinetic energy of the pulsar **[2]**
- c) The length of time (in years) for which the rotational energy associated with the pulsar stage would be able to power the luminosity of the star prior to its collapse, assuming this to have been $3000 L_{\odot}$. (Note luminosity corresponds to the total energy emitted per second by the star.) **[3]**

Comment on the result in part a), bearing in mind that pulsars often have periods in the millisecond range **[1]**.

[$1M_{\odot} = 2 \times 10^{30}$ kg; $1R_{\odot} = 6.96 \times 10^8$ m; $1L_{\odot} = 3.9 \times 10^{26}$ J/s.]

Assessed Problems 1

Richard Mason
Foundations of Data Science

Problem 1. Consider the uniform distribution

$$P(x|a) = \frac{1}{a} \quad -\frac{a}{2} \leq x \leq \frac{a}{2}$$

1. Explain the relationship between the characteristic function and the moments of a distribution. **[1]**
2. Calculate the characteristic function for the above uniform distribution. **[1]**
3. Use this to calculate the expectation value $\langle x \rangle$ and variance $\text{var}(x)$. **[2]**
4. Consider summing up N such random variables, given by

$$T = \sum_{n=1}^N x_n$$

Show that

$$P(T|N, a) = \int_{-\infty}^{\infty} \frac{dk}{2\pi} e^{ikT} \exp \left(N \log \left(\sin \frac{ka}{2} \right) - N \log \left(\frac{ka}{2} \right) \right)$$

and then as $N \rightarrow \infty$

$$P(T|N, a) \approx \int_{-\infty}^{\infty} \frac{dk}{2\pi} e^{ikT} e^{-\frac{Na^2k^2}{24}}$$

[4]

5. Evaluate the integral and show that this is consistent with the central limit theorem with the moments you found earlier. You may assume

$$\int_{-\infty}^{\infty} \frac{dx}{2\pi} e^{-a(x-ib)^2} = \frac{1}{\sqrt{4\pi a}}$$

[2]

Eigenphysics 2025-2026: Assessed Problem 1

The Schrödinger equation for the isotonic oscillator, (which is basically the radial equation for the three-dimensional harmonic oscillator), has the form

$$-\frac{d^2\psi}{dx^2} + \left[x^2 + \frac{\ell(\ell+1)}{x^2} \right] \psi = E\psi,$$

where $0 \leq x < \infty$ and $\ell = 0, 1, 2, \dots$

(a) Show that $\psi(x)$ has the asymptotic behaviour

$$\begin{aligned} \psi(x) &\sim x^{\ell+1} & \text{as } x \rightarrow 0 \\ \psi(x) &\sim e^{-x^2/2} & \text{as } x \rightarrow \infty. \end{aligned} \quad [2]$$

(b) To get rid of the asymptotics, make the substitution $\psi(x) = x^{\ell+1}e^{-x^2/2}\phi(x)$, and show that $\phi(x)$ satisfies the differential equation

$$\frac{d^2\phi}{dx^2} + 2\left(\frac{\ell+1}{x} - x\right)\frac{d\phi}{dx} + (E - 2\ell - 3)\phi = 0. \quad [3]$$

(c) If we now change the dependent variable to $z = x^2$, show that $\frac{d}{dx} = 2z^{1/2}\frac{d}{dz}$, and that the differential equation for ϕ becomes

$$4z\frac{d^2\phi}{dz^2} + (4\ell + 6 - 4z)\frac{d\phi}{dz} + (E - 2\ell - 3)\phi = 0. \quad [2]$$

(d) Expand $\phi(z)$ as a power series in z ,

$$\phi(z) = \sum_{k=0}^{\infty} a_k z^k,$$

and show that a_k satisfies the recursion relation

$$\frac{a_{k+1}}{a_k} = \frac{(4k - E + 2\ell + 3)}{4(k+1)(k + \ell + \frac{3}{2})}. \quad [3]$$